

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
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Laporan Akhir

Praktikum Jaringan Komputer

VLAN-Trunk-OSPF-Multivendor

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2026

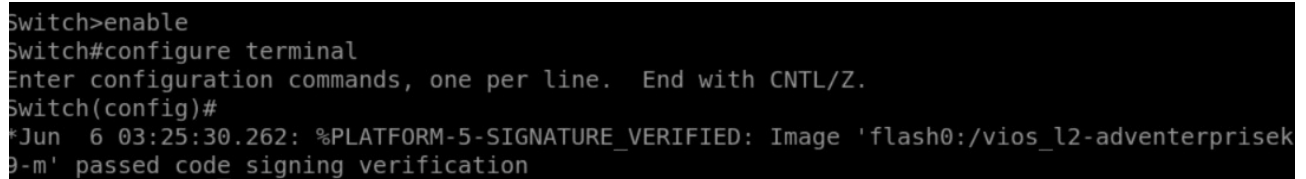
1 Langkah-Langkah Percobaan

1.1 Praktikum 1 Konfigurasi Cisco Switch

1.1.1 Langkah 1. Masuk ke terminal Cisco Switch

Masuk ke terminal Cisco Switch dengan perintah berikut:

```
Switch> enable
Switch# configure terminal
Switch(config)#
```



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
*Jun  6 03:25:30.262: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek
9-m' passed code signing verification
```

Gambar 1: Masuk ke terminal Cisco Switch.

1.1.2 Langkah 2. Membuat VLAN

Buat VLAN 10, 20, 30, dan 40 menggunakan konfigurasi berikut:

```
vlan 10
name Finance
exit
```

```
vlan 20
name HR
exit
```

```
vlan 30
name IT
exit
```

```
vlan 40
name Marketingg
exit
```

Verifikasi:

```
show vlan brief
```

```

Switch(config)#vlan 10
Switch(config-vlan)# name Finance
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)# name IT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)# name Marketingg
Switch(config-vlan)#exit
Switch(config)#

```

Gambar 2: Pembuatan VLAN.

1.1.3 Langkah 3. Pastikan VLAN sudah dibuat dan statusnya aktif

Lakukan pengecekan menggunakan perintah:

```
show vlan brief
```

Pastikan seluruh VLAN telah dibuat dan berstatus aktif.

```

Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketingg	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```

Switch#

```

Gambar 3: Hasil verifikasi VLAN.

1.1.4 Langkah 4. Konfigurasi Access Port untuk setiap VLAN

Port access harus disesuaikan dengan interface pada Cisco Switch.

```
Switch#conf t
```

```
interface gi0/2
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
exit
```

```
interface gi0/1
description PC-HR-VLAN20
switchport mode access
switchport access vlan 20
no shutdown
exit
```

```
interface gi1/1
description PC-IT-VLAN30
switchport mode access
switchport access vlan 30
no shutdown
exit
```

```
interface gi0/3
description PC-MARKETING-VLAN40
switchport mode access
switchport access vlan 40
no shutdown
exit
```

```

Switch(config)#interface gi0/2
Switch(config-if)# description PC-FINANCE-VLAN10
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/1
Switch(config-if)# description PC-HR-VLAN20
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi1/1
Switch(config-if)# description PC-IT-VLAN30
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/3
Switch(config-if)# description PC-MARKETING-VLAN40

```

Gambar 4: Konfigurasi access port.

1.1.5 Langkah 5. Konfigurasi Trunk ke Cisco Router

Konfigurasi trunk port dari switch menuju Cisco Router untuk VLAN 10, 20, 30, dan 40.

Interface	Terhubung ke	Mode	VLAN
Cisco Router Gi0/2	Trunk	10,20,30,40	heightGi0/0

Konfigurasi:

```

Switch#conf t

interface gi0/0
switchport trunk encapsulation dot1q
description TRUNK-TO-CISCO-ROUTER
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
exit

```

Verifikasi:

```
show interfaces trunk
```

```
Switch#show interfaces trunk

Port      Mode           Encapsulation  Status        Native vlan
Gi0/0     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

Gambar 5: Konfigurasi trunk port.

Jika terdapat gambar tambahan pada langkah ini:

Template Laporan Akhir/P1/img/praktikum 1/5.1.jpeg

Gambar 6: Verifikasi trunk port.

Simpan seluruh konfigurasi dengan perintah:

```
write memory
```

1.1.6 Langkah 6. Hasil yang diharapkan

Praktikum 1 selesai jika konfigurasi memenuhi berikut:

1. VLAN 10, 20, 30, dan 40 sudah ada.
2. Trunk port sudah membawa VLAN 10, 20, 30, dan 40.
3. Access port setiap VLAN sudah sesuai.

```
Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1648 bytes[OK]
Switch#
*Jun 6 03:31:58.462: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...
*Jun 6 03:31:59.191: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

Gambar 7: Hasil akhir Praktikum 1.

1.2 Praktikum 2 Konfigurasi Cisco Router

1.2.1 Langkah 1: Masuk ke Cisco Router

Masuk ke mode administrator dan mode konfigurasi global pada Cisco Router:

```
1 enable
2 configure terminal
```

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)# description TRUNK-T0-CISCO-SWITCH
Router(config-if)# no ip address
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
```

1.2.2 Langkah 2: Konfigurasi Interface Trunk ke Switch

Pilih interface yang terhubung ke Cisco Switch (pada topologi menggunakan Gig0/2):

```
1 interface gig0/2
2 description TRUNK-T0-CISCO-SWITCH
3 no ip address
4 no shutdown
5 exit
```

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)# description TRUNK-T0-CISCO-SWITCH
Router(config-if)# no ip address
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
```

1.2.3 Langkah 3: Konfigurasi Subinterface VLAN 10

```
1 interface gig0/2.10
2 description GATEWAY-VLAN10-FINANCE
3 encapsulation dot1Q 10
4 ip address 192.168.10.1 255.255.255.0
5 no shutdown
```



```
6 exit
```

```
Router(config)#interface gig0/2.10
Router(config-subif)# description GATEWAY-VLAN10-FINANCE
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)#exit
Router(config)#
```

1.2.4 Langkah 4: Konfigurasi Subinterface VLAN 20

```
1 interface gig0/2.20
2 description GATEWAY-VLAN20-HR
3 encapsulation dot1Q 20
4 ip address 192.168.20.1 255.255.255.0
5 no shutdown
6 exit
```

```
Router(config)#interface gig0/2.20
Router(config-subif)# description GATEWAY-VLAN20-HR
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

1.2.5 Langkah 5: Konfigurasi Subinterface VLAN 30

```
1 interface gig0/2.30
2 description GATEWAY-VLAN30-IT
3 encapsulation dot1Q 30
4 ip address 192.168.30.1 255.255.255.0
5 no shutdown
6 exit
```

```
Router(config)#interface gig0/2.30
Router(config-subif)# description GATEWAY-VLAN30-IT
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

1.2.6 Langkah 6: Konfigurasi Subinterface VLAN 40

```
1 interface gig0/2.40
2 description GATEWAY-VLAN40-MARKETING
3 encapsulation dot1Q 40
4 ip address 192.168.40.1 255.255.255.0
5 no shutdown
6 exit
```

```
Router(config)#interface gig0/2.40
Router(config-subif)# description GATEWAY-VLAN40-MARKETING
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

1.2.7 Langkah 7: Konfigurasi DHCP-Server untuk VLAN 10

```
1 ip dhcp excluded-address 192.168.10.1 192.168.10.10
2
3 ip dhcp pool VLAN10-FINANCE
4 network 192.168.10.0 255.255.255.0
5 default-router 192.168.10.1
6 dns-server 8.8.8.8
7 exit
```

```
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
```

1.2.8 Langkah 8: Konfigurasi DHCP-Server untuk VLAN 20

```
1 ip dhcp excluded-address 192.168.20.1 192.168.20.10
2
3 ip dhcp pool VLAN20-HR
4 network 192.168.20.0 255.255.255.0
5 default-router 192.168.20.1
6 dns-server 8.8.8.8
7 exit
```

```
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
```

1.2.9 Langkah 9: Konfigurasi IP pada VPCS (VLAN 10, 20, 30, 40)

Konfigurasi IP Static untuk PC VLAN 30 dan 40, serta request DHCP untuk VLAN 10 dan 20 di terminal VPCS masing-masing:

```
1 % Pada PC VLAN 30
2 ip 192.168.30.10/24 192.168.30.1
3
4 % Pada PC VLAN 40
```

```

5 ip 192.168.40.10/24 192.168.40.1
6
7 % Pada PC VLAN 10 & VLAN 20
8 ip dhcp

```

```

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1
VPCS> █

```

```

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

```

```

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

```

```

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

```

1.2.10 Langkah 10: Verifikasi Hasil Konfigurasi Cisco Router

Gunakan perintah berikut pada Cisco Router untuk memverifikasi status:

```

1 show ip interface brief
2 show ip dhcp pool
3 show ip dhcp binding

```

```

Router#show ip interface brief
Interface      IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
GigabitEthernet0/2    unassigned      YES unset  up                    up
GigabitEthernet0/2.10 192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20 192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30 192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40 192.168.40.1    YES manual up                    up
GigabitEthernet0/3    unassigned      YES unset  administratively down  down
GigabitEthernet0/4    unassigned      YES unset  administratively down  down
GigabitEthernet0/5    unassigned      YES unset  administratively down  down
GigabitEthernet0/6    unassigned      YES unset  administratively down  down

```

```
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 1
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.10.12      192.168.10.1 - 192.168.10.254      1

Pool VLAN20-HR :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 1
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.20.12      192.168.20.1 - 192.168.20.254      1
```

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.35    Jun 07 2026 03:48 AM    Automatic
192.168.20.11    0100.5079.6668.34    Jun 07 2026 03:48 AM    Automatic
```

1.2.11 Langkah 11: Pengujian Konektivitas Akhir dari VPCS

Lakukan pengtesan di masing-masing VPCS untuk memastikan inter-VLAN routing berjalan:

```
1 % Di PC VLAN 10
2 ip dhcp
3 show ip
4 ping 192.168.10.1
5 ping 192.168.20.11
6 ping 192.168.30.10
7 ping 192.168.40.10
8
9 % Di PC VLAN 20
10 ip dhcp
11 show ip
12 ping 192.168.20.1
13
14 % Di PC VLAN 30
15 ping 192.168.30.1
16
17 % Di PC VLAN 40
18 ping 192.168.40.1
```

```
VPCS> ip dhcp
DORA IP 192.168.10.11/24 GW 192.168.10.1
```

```
VPCS> show ip
```

```
NAME           : VPCS[1]
IP/MASK         : 192.168.10.11/24
GATEWAY         : 192.168.10.1
DNS             : 8.8.8.8
DHCP SERVER     : 192.168.10.1
DHCP LEASE      : 86394, 86400/43200/75600
MAC             : 00:50:79:66:68:35
LPORT          : 20000
RHOST:PORT      : 127.0.0.1:30000
MTU             : 1500
```

```
VPCS> ping 192.168.10.1
```

```
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.757 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.999 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=4.440 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=4.345 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.972 ms
```

```
VPCS> ip dhcp
DORA IP 192.168.20.11/24 GW 192.168.20.1
```

```
VPCS> show ip
```

```
NAME           : VPCS[1]
IP/MASK         : 192.168.20.11/24
GATEWAY         : 192.168.20.1
DNS             : 8.8.8.8
DHCP SERVER     : 192.168.20.1
DHCP LEASE      : 86396, 86400/43200/75600
MAC             : 00:50:79:66:68:34
LPORT          : 20000
RHOST:PORT      : 127.0.0.1:30000
MTU             : 1500
```

```
VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=4.708 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.600 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=4.978 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=4.815 ms
```

```
VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.845 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.117 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.824 ms
```

```
VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.420 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=4.975 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=5.594 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=4.787 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=4.788 ms
```

```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=4.723 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.717 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=6.212 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=7.725 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.862 ms
```

```
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=6.296 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.625 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=6.232 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=4.202 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.665 ms
```

```
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=6.686 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.119 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=4.822 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=4.790 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=4.790 ms
```

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.35 Jun 07 2026 03:51 AM    Automatic
192.168.20.11    0100.5079.6668.34 Jun 07 2026 03:52 AM    Automatic
```

Verifikasi kembali tabel sewa alamat IP dengan mengetikkan `show ip dhcp binding` pada terminal router.

1.3 Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

1.3.1 Langkah 1: Konfigurasi LAN Huawei

Konfigurasi IP address untuk interface lokal LAN pada router Huawei:

```
1 system-view
2 interface Ethernet1/0/2
3   description LAN-HUAWEI
4   ip address 192.168.50.1 255.255.255.0
5   undo shutdown
6   quit
7   commit
```

```
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
```

1.3.2 Langkah 2: Konfigurasi Alamat IP pada PC LAN Huawei

Atur alamat IP statis pada VPCS di sisi LAN Huawei, kemudian uji koneksi ke gateway:

```
1 ip 192.168.50.10/24 192.168.50.1
2 show ip
3 ping 192.168.50.1
```

```
VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.50.10/24
GATEWAY        : 192.168.50.1
DNS            :
MAC            : 00:50:79:66:68:36
LPORT         : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500
```

```

VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.435 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=5.208 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.702 ms

```

1.3.3 Langkah 3: Konfigurasi Alamat IP Link Cisco – Huawei (Link 1)

Konfigurasi interkoneksi network 10.10.10.0/30:

```

1 % Pada Cisco Router
2 interface gi0/3
3   description LINK1-T0-HUAWEI
4   ip address 10.10.10.1 255.255.255.252
5   no shutdown
6   exit
7
8 % Pada Huawei Router
9 system-view
10 interface Ethernet1/0/0
11   description LINK1-T0-CISCO
12   ip address 10.10.10.2 255.255.255.252
13   undo shutdown
14   quit
15 commit

```

```

Router(config)#interface gi0/3
Router(config-if)# description LINK1-T0-HUAWEI
Router(config-if)# ip address 10.10.10.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#

```

```

[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0] ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0] undo shutdown
[*HUAWEI-Ethernet1/0/0] quit
[*HUAWEI]
[*HUAWEI]commit

```

1.3.4 Langkah 4: Konfigurasi Alamat IP Link Cisco – Huawei (Link 2)

Konfigurasi interkoneksi network 10.10.11.0/30:

```

1 % Pada Cisco Router
2 interface gi0/0
3   description LINK2-T0-HUAWEI
4   ip address 10.10.11.1 255.255.255.252
5   no shutdown
6   exit
7
8 % Pada Huawei Router

```



```

9 system-view
10 interface Ethernet1/0/4
11   description LINK2-T0-CISCO
12   ip address 10.10.11.2 255.255.255.252
13   undo shutdown
14   quit
15 commit

```

```

Router(config)#interface gi0/0
Router(config-if)# description LINK2-T0-HUAWEI
Router(config-if)# ip address 10.10.11.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit

```

```

[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4] description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4] ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4] undo shutdown
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]commit

```

1.3.5 Langkah 5: Konfigurasi Alamat IP Link Huawei – MikroTik

Konfigurasi interkoneksi network 10.10.20.0/30:

```

1 % Pada Huawei Router
2 system-view
3 interface Ethernet1/0/3
4   description LINK-T0-MIKROTIK
5   ip address 10.10.20.1 255.255.255.252
6   undo shutdown
7   quit
8   commit
9
10 % Pada MikroTik Router
11 /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI

```

```

[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3] undo shutdown
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit

```

```

[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] >

```

1.3.6 Langkah 6: Konfigurasi Alamat IP Link Cisco – MikroTik

Konfigurasi interkoneksi network 10.10.30.0/30:

```

1 % Pada Cisco Router
2 interface gi0/1
3 description LINK-T0-MIKROTIK
4 ip address 10.10.30.1 255.255.255.252
5 no shutdown
6 exit
7
8 % Pada MikroTik Router
9 /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO

```

```

Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit

```

```

[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
[admin@MikroTik] >

```

1.3.7 Langkah 7: Verifikasi Status Interface di Semua Router

Pastikan status port telah berada dalam kondisi 'Up' dan alokasi alamat IP sudah tepat:

```

1 % Perintah Verifikasi pada Cisco Router
2 show ip interface brief
3
4 % Perintah Verifikasi pada Huawei Router
5 display ip interface brief
6
7 % Perintah Verifikasi pada MikroTik Router
8 /ip address print

```

```

Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.10.11.1	YES	manual	up	up
GigabitEthernet0/1	10.10.30.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/2.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/3	10.10.10.1	YES	manual	up	up
GigabitEthernet0/4	unassigned	YES	unset	administratively down	down
GigabitEthernet0/5	unassigned	YES	unset	administratively down	down
GigabitEthernet0/6	unassigned	YES	unset	administratively down	down

```

[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask      Physical  Protocol  VPN
Ethernet1/0/0            10.10.10.2/30        up        up        --
Ethernet1/0/0.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/1            unassigned           up        down      --
Ethernet1/0/1.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/2            192.168.50.1/24     up        up        --
Ethernet1/0/2.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/3            10.10.20.1/30       up        up        --
Ethernet1/0/3.4094       128.1.138.137/16    up        up        l3vpn
Ethernet1/0/4            10.10.11.2/30       up        up        --
Ethernet1/0/4.4094       128.1.138.137/16    up        up        l3vpn
---- More ----

[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
#  ADDRESS                NETWORK              INTERFACE
0  ;;; TO-HUAWEI
   10.10.20.2/30          10.10.20.0          ether1
1  ;;; TO-CISCO
   10.10.30.2/30          10.10.30.0          ether2
[admin@MikroTik] >

```

1.3.8 8. Verifikasi Ping Antar Router

1.3.9 Langkah 8: Verifikasi Uji Ping Antar-Router (Point-to-Point)

Jalankan tes konektivitas berikut untuk memastikan setiap segmen link lokal dapat terhubung:

```

1 % Dari Cisco Router
2 ping 10.10.10.2
3 ping 10.10.11.2
4 ping 10.10.30.2
5
6 % Dari Huawei Router
7 ping 10.10.10.1
8 ping 10.10.11.1
9 ping 10.10.20.2
10
11 % Dari MikroTik Router
12 ping 10.10.20.1
13 ping 10.10.30.1

```

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

```

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=8 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/3/8 ms

```

```

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

```

```

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

```

```
[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                      SIZE TTL TIME  STATUS
  0 10.10.20.1                   56 255 6ms
  1 10.10.20.1                   56 255 1ms
  2 10.10.20.1                   56 255 1ms
  3 10.10.20.1                   56 255 1ms
  4 10.10.20.1                   56 255 1ms
  5 10.10.20.1                   56 255 1ms
  6 10.10.20.1                   56 255 1ms
  7 10.10.20.1                   56 255 1ms
  8 10.10.20.1                   56 255 1ms
  9 10.10.20.1                   56 255 1ms
 10 10.10.20.1                   56 255 1ms
 11 10.10.20.1                   56 255 1ms
sent=12 received=12 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=6ms

[admin@MikroTik] > ping 10.10.30.1
  SEQ HOST                      SIZE TTL TIME  STATUS
  0 10.10.30.1                   56 255 2ms
  1 10.10.30.1                   56 255 2ms
  2 10.10.30.1                   56 255 2ms
sent=3 received=3 packet-loss=0% min-rtt=2ms avg-rtt=2ms max-rtt=2ms
```

1.4 Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1.4.1 Langkah 1: Konfigurasi Routing OSPF pada Cisco Router

Definisikan OSPF process, router-id, advertise network area 0, dan lakukan penyesuaian nilai OSPF Cost:

```
1 router ospf 1
2 router-id 1.1.1.1
3 network 10.10.10.0 0.0.0.3 area 0
4 network 10.10.11.0 0.0.0.3 area 0
5 network 10.10.30.0 0.0.0.3 area 0
6 network 192.168.10.0 0.0.0.255 area 0
7 network 192.168.20.0 0.0.0.255 area 0
8 network 192.168.30.0 0.0.0.255 area 0
9 network 192.168.40.0 0.0.0.255 area 0
10 exit
11
12 interface gi0/3
13 ip ospf cost 10
14 exit
15 interface gi0/0
16 ip ospf cost 10
17 exit
18 interface gi0/1
19 ip ospf cost 100
20 exit
```

1.4.2 Langkah 2: Konfigurasi Routing OSPF pada Huawei Router

Aktifkan routing OSPF, daftarkan network segmen area 0, dan sesuaikan parameter cost pada interface terkait:

```
1 system-view
2 ospf 1 router-id 2.2.2.2
3 area 0.0.0.0
```

```

4  network 10.10.10.0 0.0.0.3
5  network 10.10.11.0 0.0.0.3
6  network 10.10.20.0 0.0.0.3
7  network 192.168.50.0 0.0.0.255
8  quit
9  quit
10
11 interface Ethernet1/0/0
12   ospf cost 10
13   quit
14 interface Ethernet1/0/4
15   ospf cost 10
16   quit
17 interface Ethernet1/0/3
18   ospf cost 100
19   quit
20 commit

```

Enter configuration commands, one per line. End with CNTL/Z.

```

Router(config)#router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.10.10.0 0.0.0.3 area 0
Router(config-router)# network 10.10.11.0 0.0.0.3 area 0
Router(config-router)# network 10.10.30.0 0.0.0.3 area 0
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#

```

```

Router(config)#interface gi0/3
Router(config-if)# description LINK1-TO-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-TO-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/1
Router(config-if)# description LINK-TO-MIKROTIK
Router(config-if)# ip ospf cost 100
Router(config-if)#exit
Router(config)#

```

1.4.3 Langkah 3: Konfigurasi Routing OSPF pada MikroTik Router

Konfigurasi identitas router instan, definisikan area backbone, dan modifikasi nilai cost interface cadangan:

```

1 /routing ospf instance set default router-id=3.3.3.3
2 /routing ospf network add network=10.10.20.0/30 area=backbone
3 /routing ospf network add network=10.10.30.0/30 area=backbone
4 /routing ospf interface add interface=ether1 cost=100

```

```
5 /routing ospf interface add interface=ether2 cost=100
```

```
Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
  --More--
```

1.4.4 Langkah 4: Verifikasi Hubungan OSPF Neighbor

Periksa status kedekatan antar perangkat untuk menjamin relasi adjacency OSPF telah terbentuk penuh (*Full State*):

```
1 % Di Cisco Router
2 show ip ospf neighbor
3
4 % Di Huawei Router
5 display ospf peer
6
7 % Di MikroTik Router
8 /routing ospf neighbor print
```

```
[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1] area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0] quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]
```

```

[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4] description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4] ospf cost 10
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]

```

```

[~HUAWEI]display ospf peer

```

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2 Neighbors

```

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2          MTU: 1500
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m31s
  Neighbor Up Time : 2026-06-06 04:16:59
  Authentication Sequence: [ 0 ]

```

```

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1            BDR: 10.10.11.2          MTU: 1500
  Dead timer due in 30 sec
  Retrans timer interval: 5

```

---- More ----

```

public_ Routing Table : OSPF
Destinations : 9          Routes : 14

```

```

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.10.30.0/30	OSPF	10	110	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	110	D	10.10.10.1	Ethernet1/0/0
192.168.10.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
192.168.20.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
192.168.30.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0
192.168.40.0/24	OSPF	10	11	D	10.10.11.1	Ethernet1/0/4
	OSPF	10	11	D	10.10.10.1	Ethernet1/0/0

```

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.10.10.0/30	OSPF	10	10		10.10.10.2	Ethernet1/0/0
10.10.11.0/30	OSPF	10	10		10.10.11.2	Ethernet1/0/4
10.10.20.0/30	OSPF	10	100		10.10.20.1	Ethernet1/0/3
192.168.50.0/24	OSPF	10	1		192.168.50.1	Ethernet1/0/2

```

[~HUAWEI]

```



```

[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0
---- More ----

```

1.4.5 Langkah 5: Verifikasi Tabel Routing Dinamis OSPF

Periksa apakah seluruh prefix subnet remote dari vendor lain telah sukses dipelajari secara dinamis:

```

1 % Di Cisco Router
2 show ip route ospf
3
4 % Di Huawei Router
5 display ip routing-table protocol ospf
6
7 % Di MikroTik Router
8 /ip route print where ospf

```

```

admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=25s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=37s
admin@MikroTik] >

```

```

admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  10.10.10.0/30        10.10.20.1    110
      10.10.30.1
1 ADo  10.10.11.0/30        10.10.20.1    110
      10.10.30.1
2 ADo  192.168.10.0/24       10.10.30.1    110
3 ADo  192.168.20.0/24       10.10.30.1    110
4 ADo  192.168.30.0/24       10.10.30.1    110
5 ADo  192.168.40.0/24       10.10.30.1    110
6 ADo  192.168.50.0/24       10.10.20.1    110
admin@MikroTik] >

```

1.4.6 Langkah 6: Pengujian Konektivitas End-to-End Lintas Vendor

Jalankan pengujian ping dari PC di wilayah VLAN Cisco menuju PC LAN di wilayah Huawei dan sebaliknya:

```
1 % Uji coba ping dari VPCS Cisco (VLAN 10, 20, 30, dan 40)
2 ping 192.168.50.10
3
4 % Uji coba ping balik dari VPCS LAN Huawei menuju client Cisco
5 ping 192.168.10.11
6 ping 192.168.20.11
7 ping 192.168.30.10
8 ping 192.168.40.10
```

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/BDR	00:00:36	10.10.30.2	GigabitEthernet0/1
2.2.2.2	1	FULL/BDR	00:00:33	10.10.11.2	GigabitEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:36	10.10.10.2	GigabitEthernet0/3

```
Router#
```

```
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h12m08s
Neighbor Up Time : 2026-06-06 04:16:59
Authentication Sequence: [ 0 ]
```

```
Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h04m56s
Neighbor Up Time : 2026-06-06 04:24:11
Authentication Sequence: [ 0 ]
```

```
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h12m08s
Neighbor Up Time : 2026-06-06 04:16:59
Authentication Sequence: [ 0 ]
```

```
~HUAWEI]
```

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=6m16s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=6m28s
```

1.4.7 Langkah 7: Verifikasi Routing Table

```
1 % Cisco Router
2 show ip route 192.168.50.0
3 show ip route ospf
4
5 % Huawei Router
6 display ip routing-table protocol ospf
7
8 % Mikrotik
9 /ip route print where ospf
```

```
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1    FULL/BDR        00:00:36    10.10.30.2     GigabitEthernet0/1
2.2.2.2          1    FULL/BDR        00:00:33    10.10.11.2     GigabitEthernet0/0
2.2.2.2          1    FULL/BDR        00:00:36    10.10.10.2     GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:14:30 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:14:30 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:14:30 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:15:12, GigabitEthernet0/0
        [110/110] via 10.10.10.2, 00:15:12, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:15:12, GigabitEthernet0/0
        [110/11] via 10.10.10.2, 00:15:12, GigabitEthernet0/3
```

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole r
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop          Interface
-----
10.10.30.0/30      OSPF   10   110       D    10.10.11.1         Ethernet1/0/4
                  OSPF   10   110       D    10.10.10.1         Ethernet1/0/0
192.168.10.0/24    OSPF   10   11         D    10.10.11.1         Ethernet1/0/4
                  OSPF   10   11         D    10.10.10.1         Ethernet1/0/0
192.168.20.0/24    OSPF   10   11         D    10.10.11.1         Ethernet1/0/4
                  OSPF   10   11         D    10.10.10.1         Ethernet1/0/0
192.168.30.0/24    OSPF   10   11         D    10.10.11.1         Ethernet1/0/4
                  OSPF   10   11         D    10.10.10.1         Ethernet1/0/0
192.168.40.0/24    OSPF   10   11         D    10.10.11.1         Ethernet1/0/4
                  OSPF   10   11         D    10.10.10.1         Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop          Interface
-----
10.10.10.0/30      OSPF   10   10         10.10.10.2         Ethernet1/0/0
10.10.11.0/30      OSPF   10   10         10.10.11.2         Ethernet1/0/4
10.10.20.0/30      OSPF   10   100        10.10.20.1         Ethernet1/0/3
192.168.50.0/24    OSPF   10   1          192.168.50.1       Ethernet1/0/2

[~HUAWEI]
```

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#    DST-ADDRESS      PREF-SRC  GATEWAY          DISTANCE
0 ADo 10.10.10.0/30      10.10.20.1      110
      10.10.30.1
1 ADo 10.10.11.0/30      10.10.20.1      110
      10.10.30.1
2 ADo 192.168.10.0/24    10.10.30.1      110
3 ADo 192.168.20.0/24    10.10.30.1      110
4 ADo 192.168.30.0/24    10.10.30.1      110
5 ADo 192.168.40.0/24    10.10.30.1      110
6 ADo 192.168.50.0/24    10.10.20.1      110
```

1.4.8 Langkah 8: Verifikasi Koneksi end-to-end

```
1 % Dari pc vlan 10, 20, 30, 40:
2 ping 192.168.50.10
3
4 % Dari pc lan huawei ke ip vlan cisco:
5 ping 192.168.10.11
6 ping 192.168.20.11
7 ping 192.168.30.10
8 ping 192.168.40.10
```

```
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.923 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=9.919 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.368 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.184 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.126 ms
```

```

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.961 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.975 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.738 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.690 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.174 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.699 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.453 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.043 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.226 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.430 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=8.122 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.835 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.895 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.932 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.163 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.972 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=8.237 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=8.348 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=8.652 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=8.564 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=8.564 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=16.290 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=7.456 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=9.769 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=7.658 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=9.808 ms

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=14.175 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=6.334 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=7.382 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.997 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=6.109 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=8.639 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=8.495 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=8.954 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=7.646 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=9.236 ms

```

1.5 Praktikum 5 Pengujian Failover OSPF

1.5.1 Langkah 1: Verifikasi Kondisi Normal (Dua Link Utama Aktif)

Periksa tabel routing jalur spesifik dan lakukan analisis penelusuran rute data (*traceroute*):

```
1 % Di Cisco Router
2 show ip route 192.168.50.0
3
4 % Di PC VLAN Cisco
5 trace 192.168.50.10
```

```
MPCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms
```

Catatan: Pastikan paket data melompat langsung melalui link interkoneksi utama Cisco ↔ Huawei.

1.5.2 Langkah 2: Simulasi Kondisi 1 Link Cisco – Huawei Down

Matikan interface link pertama (gi0/3) pada router Cisco untuk melihat mekanisme konvergensi awal:

```
1 % Di Cisco Router
2 configure terminal
3 interface gi0/3
4 shutdown
5 exit
```

Lakukan verifikasi ulang pada rute pengiriman paket:

```
1 % Di Cisco Router
2 show ip route 192.168.50.0
3
4 % Di PC VLAN Cisco
5 trace 192.168.50.10
```

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#
*Jun  6 04:51:32.137: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down
: Interface down or detached
*Jun  6 04:51:34.105: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun  6 04:51:35.106: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
```

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:01:28 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:01:28 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
```



```
VPCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms
```

Catatan: Trafik harus dipastikan tetap memilih jalur langsung melalui link utama kedua yang masih aktif.

1.5.3 Langkah 3: Simulasi Kondisi Seluruh Link Utama Mati (Failover ke Jalur Backup)

Matikan interface link utama kedua (gi0/0) pada Cisco Router agar trafik dipaksa berpindah haluan:

```
1 % Di Cisco Router
2 configure terminal
3 interface gi0/0
4 shutdown
5 exit
```

Lakukan penelusuran rute kembali untuk membuktikan perpindahan jalur:

```
1 % Di Cisco Router
2 show ip route 192.168.50.0
3
4 % Di PC VLAN Cisco
5 trace 192.168.50.10
```

```
Router#show ip route 192.168.50.10
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.10.2 on GigabitEthernet0/3, 00:32:52 ago
  Routing Descriptor Blocks:
    * 10.10.10.2, from 2.2.2.2, 00:32:52 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
```

```
VPCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10      0.001 ms
```

Catatan: Hasil pelacakan rute harus bergeser memutar melewati rute cadangan: PC VLAN → Cisco → MikroTik → Huawei → LAN Huawei.

1.5.4 Langkah 4: Pemulihan Jaringan Kembali ke Kondisi Normal

Aktifkan kembali kedua interface link utama pada Cisco Router untuk mengembalikan struktur prioritas biaya jalur asal:

```
1 % Di Cisco Router
2 configure terminal
3 interface gi0/3
4 no shutdown
5 exit
```

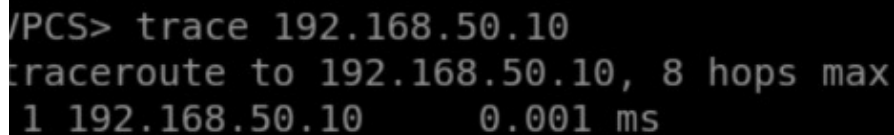
```
6 interface gi0/0
7   no shutdown
8   exit
```

Verifikasi langkah akhir untuk memastikan pemulihan jalur otomatis (*failback*):

```
1 show ip route 192.168.50.0
```

Jalankan penelusuran rute akhir dari sisi PC:

```
1 trace 192.168.50.10
```



```
/PCS> trace 192.168.50.10
traceroute to 192.168.50.10, 8 hops max
1 192.168.50.10      0.001 ms
```

2 Analisis Hasil Percobaan

Berdasarkan hasil percobaan yang telah dilakukan, implementasi arsitektur jaringan Enterprise ini berhasil berjalan dengan baik sesuai rancangan. Pembuatan VLAN di sisi Jakarta dan Surabaya terbukti efektif dalam mengisolasi broadcast domain antar-divisi demi meningkatkan keamanan dan efisiensi manajemen lalu lintas data. Sistem DHCP Relay yang dikonfigurasi pada router Jakarta juga sukses meneruskan request dari PC Client di VLAN 10 dan 20 menuju Ubuntu DHCP Server di VLAN 60, sehingga client bisa mendapatkan alokasi IP otomatis secara terpusat. Sementara itu, distribusi IP di sisi Surabaya berjalan lancar menggunakan DHCP lokal dari MikroTik untuk VLAN 30 dan IP statis manual untuk VLAN 40. Konfigurasi trunk port antar-switch dan router pun berfungsi optimal karena mampu membawa banyak VLAN ID sekaligus dalam satu koneksi fisik.

Untuk menjaga keandalan jaringan di Headquarter (HQ) Jakarta, pengujian VRRP menunjukkan bahwa Cisco Router dan MikroTik Router berhasil berkolaborasi menciptakan virtual gateway yang saling membackup. Saat Cisco bertindak sebagai Master untuk VLAN 10 dan 60, MikroTik bersiap menjadi Backup, begitu pula sebaliknya pada VLAN 20, sehingga jika salah satu router mengalami gangguan, koneksi client ke luar jaringan tidak akan terputus. Di sisi lain, meskipun kantor Jakarta dan Surabaya terpisah secara geografis dan melewati jalur publik ISP MikroTik, keduanya berhasil terhubung secara privat menggunakan GRE Tunnel yang dibangun di atas edge firewall FortiGate. Protokol routing dinamis OSPF over GRE yang diterapkan berhasil mencapai status Full neighbor dan mendistribusikan informasi routing dari kedua site secara otomatis. Berkat adanya teknik redistribute static route pada FortiGate, seluruh jaringan lokal di bawah router internal menjadi saling mengenali. Keberhasilan interkoneksi ini dibuktikan dengan suksesnya pengujian ping end-to-end antar-site serta lancarnya akses ke Web Server Nginx milik Jakarta oleh PC Client yang berada di Surabaya. Terakhir, konfigurasi Firewall Policy dan NAT/Masquerade pada FortiGate serta MikroTik ISP berjalan sesuai ekspektasi, di mana perangkat internal dari kedua site tidak hanya bisa saling bertukar data, tetapi juga memiliki akses aman untuk terhubung ke internet publik yang dibuktikan dengan suksesnya ping ke IP 8.8.8.8.

3 Hasil Tugas Modul

Ada di readme.md modul 5

4 Kesimpulan

Secara keseluruhan, percobaan ini berhasil membuktikan bahwa skenario jaringan Enterprise lintas cabang dapat berjalan dengan optimal dan memiliki keandalan yang tinggi. Integrasi VLAN dan DHCP terpusat sukses merapikan manajemen serta isolasi trafik antar-divisi, sementara implementasi VRRP di sisi Jakarta memberikan jaminan failover yang responsif sehingga jaringan tetap berjalan meskipun salah satu gateway utama tumbang. Selain itu, pemanfaatan GRE Tunnel yang dipadukan dengan OSPF over GRE terbukti menjadi solusi yang sangat efektif untuk menghubungkan site Jakarta dan Surabaya secara privat, aman, dan dinamis melalui jalur ISP. Keberhasilan seluruh konfigurasi ini divalidasi oleh lancarnya komunikasi data end-to-end, suksesnya pengujian akses internet, serta kemampuan client antar-cabang untuk saling mengakses layanan lokal seperti Web Server Nginx.

5 Lampiran

5.1 Dokumentasi saat praktikum

